International Balance Sheets: Taking Stock

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Motivation

- ***Exorbitant Privilege*** and ***Exorbitant Duty*** of the US are important empirical patterns in the field of international finance (Gourinchas and Rey 2022).

- ***Exorbitant Privilege***: US has constantly earned the positive excess return, rate of return on external asset minus that of external liabilities. (Gourinchas and Rey 2022)

- ***Exorbitant Duty***: During a crisis, wealth flows from the US to the rest of the world.

- The US case has been extensively studied.

- However, there have been fewer attempts to systematically examine the **other side of the ledger** of the US' s privilege and duty, i.e from the perspective of countries outside of US.

→ We fill this gap in the literature using the dataset on 50 countries from 1990 to 2020.

What We Do

- We analyze the properties of external returns in the ROW and how they depend on the different balance sheet structures.

- Two common views to explain the US's *Exorbitant Privilege* and *Exorbitant Duty* are ***Insurance-Provision View*** and ***Safe-Asset View*** which are closely related to the balance sheet structure. (Gourinchas and Rey 2022, Maggiori 2017, Jiang 2023, Jiang, Krishnamurthy and Lustig 2023)

- ***Insurance Provision View***: US earns excess return because of its leveraged external balance sheet.

- ***Safe Asset View***: US earns excess return because it can issue its liabilities to the ROW with low costs (special role of USD denominated assets.)

→ Given that the world is a closed economy, these theories have implications on the ROW's external returns and external balance sheets. Also, there can be a large heterogeneity within ROW itself.

Literature Review

- **Exorbitant Privilege of the US**
  - Obstfeld et al. (2005), Gourinchas and Rey (2007), Lane and Milesi-Ferretti (2007), Curcuru, Dvorak and Warnock (2008), Lane and Milesi-Ferretti (2009), Forbes (2010), Curcuru, Thomas and Warnock (2013), Atkeson, Heathcote and Perri (2022), Jiang, Richmond and Zhang (2024), Bertaut et al. (2024), Tabova and Warnock (2024)
- **Exorbitant Privilege in Other Countries**
  - Meissner and Taylor (2006), Lane and Milesi-Ferretti (2007), Habib (2010), Bracke and Schmitz (2011), Gourinchas and Rey (2014), Rogoff and Tashiro (2015), Darvas and Hüttl (2017), Adler and Garcia-Macia (2018), Milesi-Ferretti (2023), Hale and Juvenal (2023)
- **Exorbitant Duty and Reserve Currency Paradox**
  - Gourinchas and Rey (2022), Maggiori (2017), Jiang (2023), Jiang, Krishnamurthy and Lustig (2023), Devereux, Engel and Wu (2023)

Constructing the Dataset

- We assemble a comprehensive dataset that decomposes the dynamics of external balance sheets by asset class (FDI, PE, PD, OI, and FXR)
  - Macro BOP data (IMF, Haver): income balance, financial flows
  - Dataset on the currency composition of the IIP; and IIP data from External Wealth of Nations
  - **Key contribution:** IMF *Survey* to statistical authorities
- *Survey* to statistical authorities: stock-flow reconciliation tables. It includes:
  - Δ NIIP decomposed into flows, valuation due to the exchange rate, due to asset prices, and "other statistical changes"
  - Helps guide our interpretation of valuation changes
- Overall coverage: 50 countries over 1990-2020, with breakdown by asset class

External Balance Sheet, Exorbitant Privilege and Exorbitant Duty

$NFA_t - NFA_{t-1} = FA_t + V_t$ (Gourinchas and Rey 2007)

(FA_t : Financial Account, V_t : Valuation Effect)

$$FA_t = TB_t + II_t + KA_t + EO_t$$

(TB_t : Trade Balance, II_t : Income Balance, KA_t, EO_t : Capital Account, Residual (Small))

$$NFA_t - NFA_{t-1} = \Delta NFA_t \simeq TB_t + II_t + V_t = TB_t + (II_t^A - II_t^L) + (V_t^A - V_t^L)$$

$$\frac{\Delta NFA_t}{GDP_t} \simeq \frac{TB_t}{GDP_t} + \frac{(II_t^A - II_t^L) + (V_t^A - V_t^L)}{GDP_t} = \frac{TB_t}{GDP_t} + \frac{(II_t^A + V_t^A)}{A_{t-1}}\frac{A_{t-1}}{GDP_t} - \frac{(II_t^L + V_t^L)}{L_{t-1}}\frac{L_{t-1}}{GDP_t}$$

$$= \frac{TB_t}{GDP_t} + \underbrace{\overbrace{\frac{1}{2}\frac{(A_{t-1} + L_{t-1})}{GDP_t}}^{\text{Gross External Position}} \underbrace{(r_t^A - r_t^L)}_{\text{Excess Return}}}_{\text{Return Effect}} + \underbrace{\overbrace{\frac{1}{2}(r_t^A + r_t^L)}^{\text{Average Return}} \underbrace{\frac{(A_{t-1} - L_{t-1})}{GDP_t}}_{NFA_{t-1}/GDP_t}}_{\text{Position Effect}}$$

Exorbitant Privilege: $r_t^A - r_t^L > 0$ constantly, counteracting negative TB_t and negative NFA_{t-1}.

Exorbitant Duty: In times of global stress, the US sees wealth transfers to the ROW

Return over Stock Decomposition

- **Return over Stock:** $r_t^X = \frac{II_t^X + V_t^X}{X_{t-1}}$ for $X \in (A, L)$

(1) Valuation via Price and Exchange Rate

$$V_t^X = Valp_t^X + Valex_t^X$$

- *Valp* and *Valex* are observable only for some countries via the *Survey*.
- Construct $\widehat{Valex}$ via currency weight (Allen, Gautam and Juvenal 2023) ⏩ $\widehat{Valex}$
- We use $\widehat{Valex}$ as a proxy for *Valex*, and $\widehat{Valoth}$ $(= V - \widehat{Valex})$ as a proxy for *Valp*

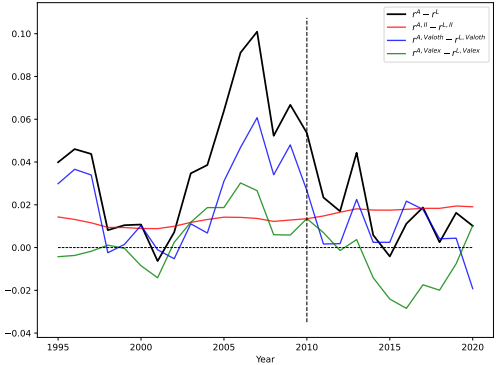
(2) (Excess) Return Decomposition

$$r_t^X = \frac{II_t^X + \widehat{Valoth}^X + \widehat{Valex}^X}{X_{t-1}} = \frac{II_t^X}{X_{t-1}} + \frac{\widehat{Valoth}^X}{X_{t-1}} + \frac{\widehat{Valex}^X}{X_{t-1}} = r^{X,II} - r^{X,Valoth} - r^{X,Valex} \text{ for } X \in (A, L)$$

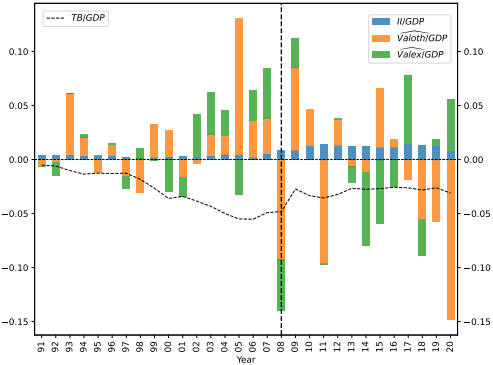
$$\rightarrow \; r_t^A - r_t^L = (r_t^{A,II} - r_t^{L,II}) + (r_t^{A,Valoth} - r_t^{L,Valoth}) + (r_t^{A,Valex} - r_t^{L,Valex})$$

Exorbitant Privilege and Duty of the US

Estimates



(a) Excess Return ($r^A - r^L$), MA(5)



(b) External Return vs Trade Balance

Figure 1. US Exorbitant Privilege and Exorbitant Duty

The Other Side of the Ledger of the US Exorbitant Privilege

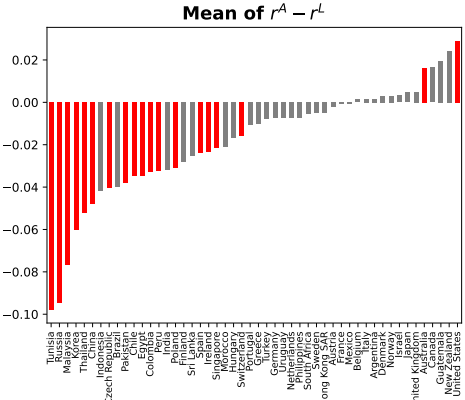
- **Excess Return of Other Countries:** $r^A - r^L$

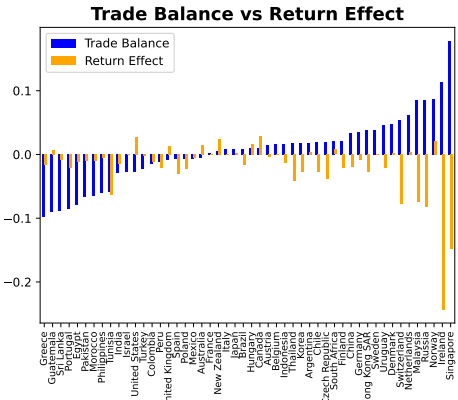
- Are there winners other than the US?
- How heterogeneous are they?
- Are they quantitatively important?

$$\frac{\Delta NFA_t}{GDP_t} \simeq \frac{TB_t}{GDP_t} + \frac{(II_t^A - II_t^L) + (V_t^A - V_t^L)}{GDP_t} = \frac{TB_t}{GDP_t} + \frac{(II_t^A + V_t^A)}{A_{t-1}}\frac{A_{t-1}}{GDP_t} - \frac{(II_t^L + V_t^L)}{L_{t-1}}\frac{L_{t-1}}{GDP_t}$$

$$= \frac{TB_t}{GDP_t} + \underbrace{\overbrace{\frac{1}{2}\frac{(A_{t-1}+L_{t-1})}{GDP_t}}^{\text{Gross External Position}}\underbrace{(r_t^A - r_t^L)}_{\text{Excess Return}}}_{\text{Return Effect}} + \underbrace{\overbrace{\frac{1}{2}(r_t^A + r_t^L)}^{\text{Average Return}}\underbrace{\frac{(A_{t-1}-L_{t-1})}{GDP_t}}_{NFA_{t-1}/GDP_t}}_{\text{Position Effect}}$$

Are There non-US Winners? No, But Substantial Heterogeneity: 1990-2020





→ Red bar plot implies that the excess is significantly different from 0.

1991-2006 2010-2019

Balance Sheet Decomposition: Asset-Class and Currency Composition

- **Asset-Class Decomposition**

  $A_t = A_t^{peq} + A_t^{pd} + A_t^{deq} + A_t^{dd} + A_t^{oi} + A_t^{fx}$ (Same for Liabilities Side)

  peq: Portfolio Investment Equity, *pd*: Portfolio Investment Debt

  deq: FDI Equity, *dd*: FDI Debt

  oi: Other Investment (mostly bank loans), *fx*: FX reserve

- **Currency Decomposition**

  $A_t = A_t^{dc} + A_t^{nonusdfc} + A_t^{usd}$ (Same for Liabilities Side)

  dc : Domestic currency

  nonusdfc : Foreign currency other than USD

  usd : US Dollar

Driver of Return Differentials: Balance Sheet Structure

- **Return Differentials and Balance Sheets**

- Can we explain the return differentials ($r^A - r^L$) across ROW?

1) Insurance-Provision View: $\mu^{eq} = \frac{A_t^{peq} + A_t^{deq} + A_t^{dd}}{A_t} - \frac{L_t^{peq} + L_t^{deq} + L_t^{dd}}{L_t}$

2) Safe Asset View: $\mu^{usd} = \frac{A_t^{usd}}{A_t} - \frac{L_t^{usd}}{L_t}$

$\rightarrow$ $y_{it} = \beta_1 \mu_{it-1}^{eq} + \beta_2 \mu_{it-1}^{usd} + \epsilon_{it}$ where $y_{it} \in (r_{it}^A - r_{it}^L, r_{it}^{A,II} - r_{it}^{L,II}, r_{it}^{A,Valoth} - r_{it}^{L,Valoth}, r_{it}^{A,Valex} - r_{it}^{L,Valex})$

- Document systematic correlations

- We examine the variations across 1) non-US countries, 2) ADV, and 3) EME.

- We examine any change between before GFC and after GFC.

Result 1: Systematic Patterns - Balance Sheet Variables and Excess Returns

All non-US Countries	$r^A - r^L$	$r^{A,II} - r^{L,II}$	$r^{A,Valoth} - r^{L,Valoth}$	$r^{A,Valex} - r^{L,Valex}$
No. Observations	1360	1360	1380	1387
R-squared	0.07	0.374	0.02	0.011
$\mu^{n,eq}$	0.078***	0.025***	0.050**	0.003
	(4.123)	(5.193)	(2.226)	(0.206)
$\mu^{n,usd}$	-0.028***	-0.029***	-0.040***	0.040***
	(-2.821)	(-16.303)	(-3.329)	(3.218)

Advanced Economies	$r^A - r^L$	$r^{A,II} - r^{L,II}$	$r^{A,Valoth} - r^{L,Valoth}$	$r^{A,Valex} - r^{L,Valex}$
No. Observations	720	720	740	747
R-squared	0.034	0.134	0.014	0.002
$\mu^{n,eq}$	0.064***	0.015***	0.043**	0.005
	(3.534)	(4.243)	(2.024)	(0.637)
$\mu^{n,usd}$	-0.015*	-0.011***	-0.006	0.003
	(-1.798)	(-10.119)	(-0.767)	(0.407)

Emerging Economies	$r^A - r^L$	$r^{A,II} - r^{L,II}$	$r^{A,Valoth} - r^{L,Valoth}$	$r^{A,Valex} - r^{L,Valex}$
No. Observations	640	640	640	640
R-squared	0.086	0.501	0.027	0.022
$\mu^{n,eq}$	0.062**	-0.006	-0.022	0.090**
	(2.030)	(-0.842)	(-0.438)	(2.063)
$\mu^{n,usd}$	-0.045***	-0.055***	-0.095***	0.105***
	(-2.375)	(-14.807)	(-3.067)	(3.438)

- **Regressions for 1990-2020**

1. Strong role of μ^{eq} and μ^{usd}
   - $\mu^{eq} > 0$: Leverage
   - $\mu^{usd} < 0$: Low Returns of US liabilities
2. Large R^2 for $r^{A,II} - r^{L,II}$
3. Asymmetry between in ADV and EME
   - ADV: no role of USD in $Valoth, Valex$
   - EME: strong role of USD in $Valoth, Valex$
   - FX Stability/Types of USD assets

Result 2: Declining Role of the USD in Explaining Excess Returns

	Before GFC (1990-2006)			After GFC (2010-2020)		
	Non-US	ADV	EME	Non-US	ADV	EME
			$r^A - r^L$			
No. Observations	702	370	332	470	250	220
R-squared	0.039	0.032	0.048	0.197	0.098	0.227
$\mu^{n,eq}$	0.049*	0.059**	0.007	0.124***	0.073***	0.141***
	(1.880)	(2.188)	(0.148)	(6.062)	(4.041)	(5.817)
$\mu^{n,usd}$	-0.043***	-0.028*	-0.075***	-0.010	0.003	-0.010
	(-2.861)	(-1.874)	(-3.072)	(-0.912)	(0.447)	(-0.535)
			$r^{A,II} - r^{L,II}$			
R-squared	0.267	0.139	0.393	0.581	0.205	0.69
$\mu^{n,eq}$	0.007*	0.006	-0.027***	0.055***	0.029***	0.035***
	(1.733)	(1.199)	(-4.168)	(19.740)	(18.536)	(10.887)
$\mu^{n,usd}$	-0.033***	-0.016***	-0.065***	-0.023***	-0.007***	-0.040***
	(-12.001)	(-10.332)	(-11.137)	(-18.279)	(-15.757)	(-16.200)
			$r^{A,Valoth} - r^{L,Valoth}$			
R-squared	0.015	0.019	0.026	0.14	0.025	0.187
$\mu^{n,eq}$	0.010	0.048	-0.149*	0.103***	0.028**	0.124***
	(0.297)	(1.540)	(-1.819)	(4.889)	(1.979)	(3.598)
$\mu^{n,usd}$	-0.075***	-0.020	-0.200***	-0.010	0.009	-0.012
	(-3.110)	(-1.292)	(-3.202)	(-1.269)	(1.121)	(-1.002)
			$r^{A,Valex} - r^{L,Valex}$			
R-squared	0.01	0.012	0.025	0.071	0.013	0.102
$\mu^{n,eq}$	0.032	0.004	0.182***	-0.033*	0.016**	-0.019
	(1.377)	(0.421)	(2.520)	(-1.818)	(2.042)	(-0.428)
$\mu^{n,usd}$	0.067***	0.009	0.190***	0.023	0.001	0.042
	(3.140)	(0.773)	(3.206)	(1.629)	(0.068)	(1.530)

- **Regressions for before GFC vs after GFC**

1. Role of $\mu^{eq} \uparrow$
2. Role of $\mu^{usd} \downarrow$
3. β of μ^{usd} for r^{II} weaker
   - $|\beta^{usd}|$ got smaller with high t stat
   - Weakening of special role of the USD?
3. ADV and EME become more similar
   - No special role of *Valex*
   - strong role of μ^{eq} in *Valoth*
   - Similar within-heterogeneity dynamics

The Other Side of the Ledger of the US's Exorbitant Duty

- **Exorbitant Duty**

- Gourinchas and Rey (2022) showed that US net foreign asset deteriorate during the time of global stress. (i.e., US transfers its wealth to the ROW.) Result

- $NFA/GDP_t = \alpha + \beta_t VIX_t + \epsilon_t$ and $V_t/GDP_t = \alpha + \beta_t VIX_t + \epsilon_t$

- **The Other Side of the Ledger**

- Regressions to examine whether the ROW benefits in times of global stress.

  $\frac{II_{it}+V_{it}}{GDP_{it}} = \beta UI_t + \epsilon_{it}$

  UI_t : (1) Global Financial Cycles Factor, (2) VIX, **(3) BAA-Treasury 10Y Spread**, (4) Policy Uncertainty, (5) Financial Stress Index, (6) GZ Spread, (7) World Uncertainty Index UI

$\rightarrow$ $\beta > 0$ are estimated positive over different samples and time horizons. Back

Exorbitant Duty and External Balance Sheets

- **Heterogeneity in the flip side of the US's Exorbitant Duty and Balance Sheet**

- Regressions to examine patterns between Exorbitant Duty and the balance sheet.

$$\frac{II_{it}+V_{it}}{GDP_{it}} = \beta_1 UI_t + \beta_2 \mu_{it}^{eq} + \beta_3 \mu_{it}^{usd} + \beta_{eq} \mu_{it}^{eq} \times UI_t + \beta_{usd} \mu_{it}^{usd} \times UI_t + \beta_4 \frac{NFA}{GDP}_{it} + \epsilon_{it}$$

$$\frac{Valoth_{it}}{GDP_{it}} = \beta_1 UI_t + \beta_2 \mu_{it}^{eq} + \beta_3 \mu_{it}^{usd} + \beta_{eq} \mu_{it}^{eq} \times UI_t + \beta_{usd} \mu_{it}^{usd} \times UI_t + \beta_4 \frac{NFA}{GDP}_{it} + \epsilon_{it}$$

$$\frac{Valex_{it}}{GDP_{it}} = \beta_1 UI_t + \beta_2 \mu_{it}^{eq} + \beta_3 \mu_{it}^{usd} + \beta_{eq} \mu_{it}^{eq} \times UI_t + \beta_{usd} \mu_{it}^{usd} \times UI_t + \beta_4 \frac{NFA}{GDP}_{it} + \epsilon_{it}$$

$\beta_{eq}, \beta_{usd} \rightarrow$ Exorbitant Duty and Balance Sheet?

1) Check interaction between balance sheet variables and UI.
2) Control $\frac{NFA}{GDP}$ Reg Specification Motivation
3) Exclude outliers, the countries with large NFA/GDP.[1]

[1] Financial centers including Belgium, Hong Kong S.A.R., Ireland, Netherlands, Singapore, Switzerland, the UK and oil-producing country with large pension fund including Norway.

The Other Side of the Exorbitant Duty Ledger: 1990-2020

1) $\beta_2 > 0, \ \beta_3 < 0$

2) Little Systematic Pattern over the Full Sample.

	$(II + Valoth + Valex)/GDP$			$Valoth/GDP$			$Valex/GDP$		
	nonUS	ADV	EME	nonUS	ADV	EME	nonUS	ADV	EME
No. Observations	1356	496	638	1376	505	638	1383	505	638
Cov. Est.	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered
R-squared	0.124	0.063	0.234	0.009	0.016	0.025	0.002	0.024	0.018
$\Delta ln(BAA)$	-0.030	0.069	-0.018	-0.005	0.100	0.036	-0.019	-0.03***	-0.064
	(-0.799)	(1.102)	(-0.353)	(-0.119)	(1.62)	(0.362)	(-0.583)	(-2.635)	(-0.598)
$\mu^{n,eq}$	0.059***	0.027	0.041***	0.038*	0.011	0.147***	-0.006	0.004	-0.088
	(4.326)	(0.993)	(2.402)	(1.795)	(0.4)	(2.392)	(-0.391)	(0.389)	(-1.436)
$\mu^{n,usd}$	-0.014	-0.019	-0.042***	0.023	-0.005	-0.001	-0.009	-0.003	0.007
	(-1.07)	(-1.462)	(-2.741)	(1.505)	(-0.322)	(-0.044)	(-0.754)	(-0.267)	(0.316)
$\Delta ln(BAA)$:$\mu^{n,eq}$	0.017	-0.117	0.049	-0.054	-0.176	-0.08	0.087	0.083	0.133
	(0.211)	(-1.194)	(0.521)	(-0.499)	(-1.36)	(-0.54)	(1.324)	(1.222)	(1.061)
$\Delta ln(BAA)$:$\mu^{n,usd}$	0.209**	-0.054	0.217***	0.078	-0.168	-0.001	0.115	0.108*	0.222
	(2.178)	(-0.476)	(2.381)	(0.881)	(-1.23)	(-0.004)	(1.591)	(1.815)	(1.445)
NFA/GDP	0.055***	0.042*	0.018	0.022	0.001	-0.113***	0.006	0.001	0.098***
	(2.505)	(1.855)	(1.219)	(1.107)	(0.021)	(-3.173)	(0.86)	(0.11)	(3.026)

The Other Side of the Exorbitant Duty Ledger: 2007-2009

1) For *Valoth*: $\beta_{eq} < 0, \beta_{usd} > 0$ and For *Valex*: $\beta_{eq} > 0, \beta_{usd} > 0$

2) Heterogeneity between ADV and EME of β_{usd} for Valoth $\rightarrow$ Different USD Asset types?

	$(II + Valoth + Valex)/GDP$			$Valoth/GDP$			$Valex/GDP$		
	nonUS	ADV	EME	nonUS	ADV	EME	nonUS	ADV	EME
No. Observations	141	51	66	141	51	66	141	51	66
Cov. Est.	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered	Clustered
R-squared	0.13	0.339	0.509	0.076	0.204	0.298	0.105	0.186	0.452
$\Delta ln(BAA)$	-0.047*	0.173***	-0.082***	0.029	0.198***	0.035*	-0.038***	-0.053***	-0.053***
	(-1.907)	(2.942)	(-4.335)	(1.388)	(3.971)	(1.889)	(-7.287)	(-10.79)	(-7.079)
$\mu^{n,eq}$	0.021**	0.044	-0.073	0.076***	0.072	0.033	-0.068***	-0.071***	-0.047*
	(2.159)	(0.623)	(-1.16)	(2.717)	(0.836)	(0.431)	(-11.286)	(-4.27)	(-1.81)
$\mu^{n,usd}$	-0.147***	-0.124***	-0.208***	-0.058*	-0.07*	-0.097***	-0.053***	-0.053***	-0.037*
	(-3.202)	(-3.11)	(-10.903)	(-1.813)	(-1.839)	(-3.494)	(-9.893)	(-2.963)	(-1.888)
$\Delta ln(BAA)$:$\mu^{n,eq}$	0.02	-0.158***	0.048	-0.306***	-0.437***	-0.192*	0.332***	0.336***	0.29***
	(1.56)	(-2.751)	(0.562)	(-16.332)	(-5.684)	(-1.859)	(27.727)	(13.835)	(7.993)
$\Delta ln(BAA)$:$\mu^{n,usd}$	0.406***	0.023	0.504***	0.059	-0.184*	0.098**	0.277***	0.255***	0.327***
	(3.923)	(0.189)	(13.521)	(0.754)	(-1.878)	(2.039)	(19.374)	(8.236)	(12.879)
NFA/GDP	0.022	0.105	0.038	0.016	0.053	-0.014	-0.015	-0.016***	0.019
	(0.46)	(1.144)	(1.499)	(0.43)	(0.584)	(-0.39)	(-0.768)	(-6.132)	(0.877)

Conclusion

Conclusion

- **Conclusion**

- We provide the new empirical evidence on the other side of the ledger of the US's exorbitant privilege and duty.
- There is a large heterogeneity in excess return across countries and its quantitative importance is comparable to the trade balance.
- Excess returns and balance sheet structures (equity share and USD share) show strong cross-country association.

→ More leverage implies higher returns & More USD assets implies lower returns.

- The ROW's external position improves in average during the time of global stress, and balance sheet asymmetries across countries matters for heterogeneous impacts of exorbitant duty across countries.

→ More leveraged → Less balance sheet gains due to the loss in *Valoth*

→ More USD → More balance sheet gains due to gain in *Valoth* (EME only) and *Valex*.

Appendix

How to calculate $\widehat{VALEX}$ Back

- For country i, following Allen, Gautam and Juvenal (2023), define a financially weighted exchange rate index based on currency weight w_c,

$$I_{t+1} = I_t * (1 + \sum w_{c,t} \% \Delta E_{ic,t})$$

$E_{ic,t}$: Nominal Exchange Rate of i against c, $\quad w_{c,t} = \frac{A_t^c}{A_t}$

- Then, $\widehat{VALEX}_t$ is calculated as:

$$\widehat{VALEX}_t = \% \Delta I_{t+1} A_t$$

$\rightarrow$ This gives us $\widehat{VALEX}_t$

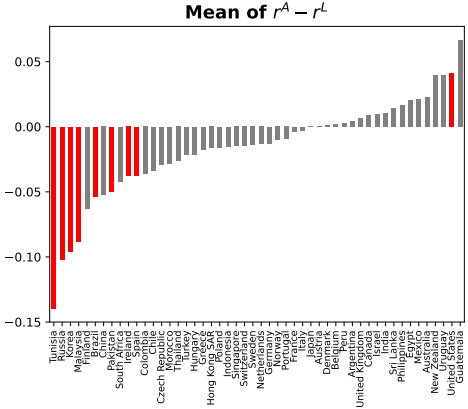
Measures of US Exorbitant Privilege in the Previous Literature

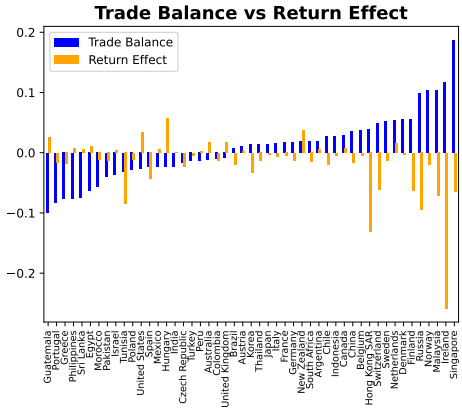
Back

	Period 1952:1–2011:4	1952:1–1972:4	1973:1–2011:4
(a) OC_{t+1} allocated to flows	1.6	0.8	2.0
(b) OC_{t+1} allocated to flows (except for FDI)	2.1	0.8	2.8
(c) OC_{t+1} allocated to valuations	2.7	0.8	3.8
Previous estimates			
(d) Initial Gourinchas and Rey (2007a) on 1952–2004	2.1		
(e) Curcuru et al. (2008b) on 1994–2005	0.72		
(f) Forbes (2010) on 2002–2008	6.9		
(g) Lane and Milesi-Ferretti (2009) on 1980–2004	3.9		
(h) Obstfeld and Rogoff (2005) on 1983–2003	3.1		

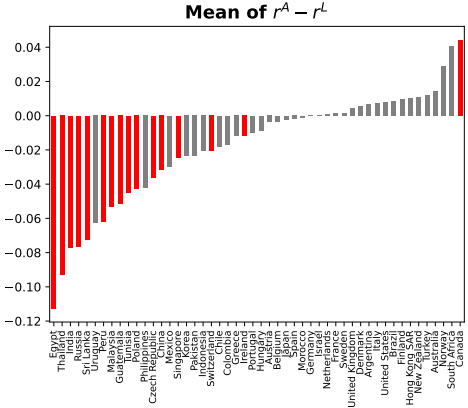
Table 10 from Gourinchas and Rey (2014)

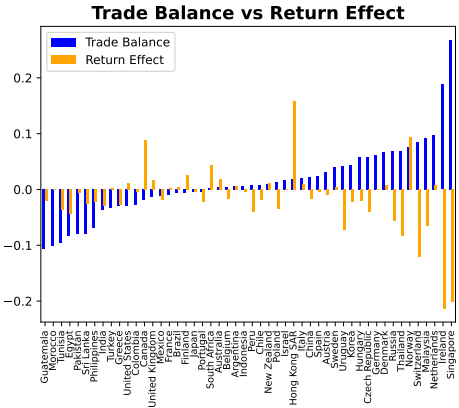
Substantial Heterogeneity in Excess Returns (1991-2006) Back





Substantial Heterogeneity in Excess Returns (2010-2019) Back





Countries with Highest Excess Return: $r^A - r^L$

Back

Rank	1991-2020	$r^A - r^L$	$p - val$	1991-2006	$r^A - r^L$	$p - val$	2007-2008	$r^A - r^L$	2010-2019	$r^A - r^L$	$p - val$
1	**United States**	0.029*	0.025	Guatemala	0.066	0.276	South Africa	0.078	Canada	0.035	0.064
2	**New Zealand**	0.024	0.219	**United States**	0.041*	0.019	Turkey	0.074	Norway	0.028	0.095
3	Guatemala	0.019	0.561	Uruguay	0.039	0.099	Greece	0.067	**United States**	0.024	0.247
4	Canada	0.016	0.133	**New Zealand**	0.039	0.288	Guatemala	0.056	South Africa	0.019	0.505
5	**Australia**	0.016*	0.029	**Australia**	0.023*	0.051	Korea	0.053	**Australia**	0.016	0.104
6	United Kingdom	0.004	0.324	Mexico	0.021	0.510	Japan	0.043	Denmark	0.015	0.135
7	Japan	0.004	0.757	Egypt	0.020	0.307	Argentina	0.040	Finland	0.014	0.059
8	Israel	0.003	0.830	Philippines	0.016	0.506	Brazil	0.033	**New Zealand**	0.013	0.146
9	Norway	0.003	0.765	Sri Lanka	0.014	0.582	Israel	0.025	Italy	0.009	0.132
10	Denmark	0.003	0.659	India	0.010	0.773	United Kingdom	0.025	Turkey	0.004	0.900

Table A.1. Excess Return Countries in Different Periods

Countries with Lowest Excess Return: $r^A - r^L$ Back

Rank	1991-2020	$r^A - r^L$	$p - val$	1991-2006	$r^A - r^L$	$p - val$	2007-2008	$r^A - r^L$	2010-2019	$r^A - r^L$	$p - val$
1	Tunisia	-0.097*	0.0000	Tunisia	-0.139*	0.0000	Indonesia	-0.215	Egypt	-0.108*	0.0000
2	**Russia**	-0.094*	0.0015	**Russia**	-0.101*	0.0255	Chile	-0.136	**Russia**	-0.108*	0.0083
3	**Malaysia**	-0.076*	0.0008	Korea	-0.095*	0.0137	Pakistan	-0.130	Thailand	-0.107*	0.0015
4	Korea	-0.060*	0.0123	**Malaysia**	-0.087*	0.0256	Czech Republic	-0.118	Sri Lanka	-0.095*	0.0028
5	Thailand	-0.052*	0.0437	Finland	-0.062	0.2616	Peru	-0.114	India	-0.080*	0.0139
6	China	-0.047*	0.0161	Brazil	-0.053*	0.0088	Tunisia	-0.097	Peru	-0.073*	0.0005
7	Indonesia	-0.041	0.1259	China	-0.052	0.0961	India	-0.083	**Malaysia**	-0.068*	0.0073
8	Czech Republic	-0.040*	0.0003	Pakistan	-0.049*	0.0318	**Malaysia**	-0.077	Uruguay	-0.065	0.0948
9	Brazil	-0.040	0.1573	South Africa	-0.042	0.0670	Singapore	-0.077	Indonesia	-0.058	0.3204
10	Pakistan	-0.038*	0.0192	Ireland	-0.037*	0.0246	Poland	-0.072	Guatemala	-0.052*	0.0000

Table A.2. Low Excess Return Countries in Different Periods

Countries with Highest Excess Yield: $r^{A,ll} - r^{L,ll}$ Back

Rank	1991-2020	$r^{A,ll} - r^{L,ll}$	$p - val$	1991-2006	$r^{A,ll} - r^{L,ll}$	$p - val$	2007-2008	$r^{A,ll} - r^{L,ll}$	2010-2019	$r^{A,ll} - r^{L,ll}$	$p - val$
1	**Japan**	0.0168*	0.0000	Egypt	0.0167*	0.0023	Egypt	0.0240	**United States**	0.0185*	0.0000
2	**United States**	0.0145*	0.0000	**Japan**	0.0155*	0.0000	**Japan**	0.0203	**Japan**	0.0176*	0.0000
3	**Denmark**	0.0085*	0.0027	**United States**	0.0123*	0.0000	Sweden	0.0127	Germany	0.0087*	0.0000
4	**Switzerland**	0.0059*	0.0001	**Switzerland**	0.0115*	0.0000	Sri Lanka	0.0118	**Denmark**	0.0084*	0.0000
5	**France**	0.0047*	0.0000	**Denmark**	0.0087	0.0977	India	0.0112	Sweden	0.0067*	0.0000
6	Sweden	0.0045*	0.0363	Sri Lanka	0.0072*	0.0115	**United States**	0.0104	**France**	0.0064*	0.0000
7	Germany	0.0011	0.4121	Uruguay	0.0037	0.4922	Philippines	0.0103	Korea	0.0054*	0.0244
8	Netherlands	0.0011	0.2771	**France**	0.0033*	0.0060	Morocco	0.0085	Spain	0.0038	0.1670
9	Turkey	0.0005	0.6951	Philippines	0.0022	0.6938	**Denmark**	0.0077	Italy	0.0017*	0.0395
10	United Kingdom	0.0000	0.8856	Netherlands	0.0020*	0.2687	**France**	0.0071	**Switzerland**	0.0014	0.2596

Table A.3. High Excess Yield Countries in Different Periods

Countries with Lowest Excess Yield: $r^{A,ll} - r^{L,ll}$ Back

Rank	1991-2020	$r^{A,ll} - r^{L,ll}$	$p - val$	1991-2006	$r^{A,ll} - r^{L,ll}$	$p - val$	2007-2008	$r^{A,ll} - r^{L,ll}$	2010-2019	$r^{A,ll} - r^{L,ll}$	$p - val$
1	**Peru**	-0.050*	0.0000	**Chile**	-0.046*	0.0000	**Chile**	-0.114	**Peru**	-0.072*	0.0000
2	**Chile**	-0.047*	0.0000	Colombia	-0.040*	0.0000	**Peru**	-0.107	**Argentina**	-0.067*	0.0000
3	**Argentina**	-0.046*	0.0000	**Russia**	-0.038*	0.0000	Colombia	-0.071	Thailand	-0.050*	0.0000
4	**Russia**	-0.041*	0.0000	Pakistan	-0.037*	0.0000	**Argentina**	-0.050	**Russia**	-0.049*	0.0000
5	Colombia	-0.039*	0.0000	Brazil	-0.036*	0.0000	Indonesia	-0.038	**Czech Republic**	-0.044*	0.0000
6	**Czech Republic**	-0.037*	0.0000	Malaysia	-0.034*	0.0000	Guatemala	-0.038	Guatemala	-0.043*	0.0000
7	Guatemala	-0.032*	0.0000	South Africa	-0.032*	0.0000	Poland	-0.038	Indonesia	-0.041*	0.0000
8	Brazil	-0.032*	0.0000	**Argentina**	-0.032*	0.0000	Thailand	-0.036	Uruguay	-0.040*	0.0000
9	Pakistan	-0.030*	0.0000	**Czech Republic**	-0.031*	0.0000	**Czech Republic**	-0.035	Egypt	-0.039*	0.0000
10	Malaysia	-0.028*	0.0000	**Peru**	-0.031*	0.0029	Uruguay	-0.030	**Chile**	-0.037*	0.0001

Table A.4. Low Excess Yield Countries in Different Periods

Countries with Highest Excess *Valoth* Return: $r^{A,Valoth} - r^{L,Valoth}$

Back

Rank	1991-2020	$r^{A,Valo} - r^{L,Valo}$	$p - val$	1991-2006	$r^{A,Valo} - r^{L,Valo}$	$p - val$	2007-2008	$r^{A,Valo} - r^{L,Valo}$	2010-2019	$r^{A,Valo} - r^{L,Valo}$	$p - val$
1	Guatemala	0.0459	0.1730	Guatemala	0.0799	0.2043	Japan	0.0935	**Canada**	0.0399*	0.0148
2	**New Zealand**	0.0401*	0.0493	**New Zealand**	0.0573	0.1218	Guatemala	0.0916	**New Zealand**	0.0304*	0.0158
3	**Australia**	0.0282*	0.0000	**Australia**	0.0325*	0.0024	Greece	0.0830	**Australia**	0.0282*	0.0105
4	**Canada**	0.0244*	0.0268	**United States**	0.0251	0.0963	China	0.0806	Norway	0.0238	0.1210
5	**United States**	0.0135	0.1983	**Canada**	0.0230	0.1014	Argentina	0.0706	Finland	0.0133	0.0557
6	**Israel**	0.0100	0.5001	Peru	0.0195	0.1345	**Israel**	0.0508	Hong Kong SAR	0.0124	0.2507
7	Peru	0.0083	0.3358	Uruguay	0.0162	0.4844	Thailand	0.0405	**United States**	0.0124	0.4045
8	Norway	0.0074	0.5042	Czech Republic	0.0102	0.3359	Switzerland	0.0257	**Israel**	0.0120	0.6132
9	Hong Kong SAR	0.0048	0.7216	**Israel**	0.0060	0.7886	Turkey	0.0245	South Africa	0.0105	0.4235
10	Czech Republic	0.0017	0.8000	Sri Lanka	0.0027	0.9165	Belgium	0.0156	Italy	0.0064	0.2949

Table A.5. High Excess *Valoth* Return Countries in Different Periods

Countries with Lowest Excess *Valoth* Return: $r^{A,Valoth} - r^{L,Valoth}$

Back

Rank	1991-2020	$r^{A,Valo} - r^{L,Valo}$	$p - val$	1991-2006	$r^{A,Valo} - r^{L,Valo}$	$p - val$	2007-2008	$r^{A,Valo} - r^{L,Valo}$	2010-2019	$r^{A,Valo} - r^{L,Valo}$	$p - val$
1	**Brazil**	-0.337	0.0594	**Brazil**	-0.591	0.0773	Indonesia	-0.220	Egypt	-0.162*	0.0572
2	**Russia**	-0.131*	0.0011	**Russia**	-0.170*	0.0283	India	-0.148	**Russia**	-0.102*	0.0060
3	**Tunisia**	-0.105*	0.0000	**Tunisia**	-0.147*	0.0001	Pakistan	-0.148	India	-0.089*	0.0001
4	**Turkey**	-0.089*	0.0000	**Turkey**	-0.125*	0.0000	Chile	-0.083	Sri Lanka	-0.085*	0.0078
5	Egypt	-0.079*	0.0233	Korea	-0.085*	0.0080	Tunisia	-0.075	**Tunisia**	-0.061*	0.0070
6	Korea	-0.061*	0.0010	**Malaysia**	-0.072	0.0752	Canada	-0.075	Argentina	-0.053*	0.0234
7	**Malaysia**	-0.059*	0.0076	Finland	-0.063	0.2834	Russia	-0.070	**Brazil**	-0.053	0.1775
8	India	-0.055*	0.0208	South Africa	-0.056	0.1586	Czech Republic	-0.066	**Malaysia**	-0.050*	0.0044
9	Indonesia	-0.038	0.1240	China	-0.040	0.2377	Korea	-0.064	Thailand	-0.045*	0.0150
10	Thailand	-0.029	0.2216	Spain	-0.038*	0.0014	Egypt	-0.059	**Turkey**	-0.043	0.0948

Table A.6. Low Excess *Valoth* Return Countries in Different Periods

Countries with Highest Excess *Valex* Return: $r^{A,Valex} - r^{L,Valex}$

Back

Rank	1991-2020	$r^{A,Vale} - r^{L,Vale}$	$p - val$	1991-2006	$r^{A,Vale} - r^{L,Vale}$	$p - val$	2007-2008	$r^{A,Vale} - r^{L,Vale}$	2010-2019	$r^{A,Vale} - r^{L,Vale}$	$p - val$
1	**Brazil**	0.3305	0.0633	**Brazil**	0.5751	0.0846	**South Africa**	0.1173	**Argentina**	0.1149*	0.0050
2	**Turkey**	0.0812*	0.0000	**Russia**	0.1072	0.1094	Korea	0.1145	**Egypt**	0.0933	0.2511
3	**Russia**	0.0789*	0.0259	**Turkey**	0.1025*	0.0002	Mexico	0.0736	**Turkey**	0.0492*	0.0147
4	**Argentina**	0.0713*	0.0118	**Argentina**	0.0475	0.2976	**Brazil**	0.0639	**Russia**	0.0439	0.2558
5	**Egypt**	0.0497	0.1004	**South Africa**	0.0467	0.1580	Chile	0.0601	**Brazil**	0.0346	0.2690
6	**South Africa**	0.0470*	0.0401	Mexico	0.0416	0.0822	**India**	0.0540	Tunisia	0.0338*	0.0021
7	Mexico	0.0352*	0.0154	**Egypt**	0.0314	0.0703	United Kingdom	0.0521	**South Africa**	0.0325	0.3871
8	**India**	0.0288*	0.0082	**India**	0.0286*	0.0222	**Russia**	0.0517	Pakistan	0.0243*	0.0195
9	**Colobmia**	0.0241*	0.0264	**Colobmia**	0.0277*	0.0263	**Turkey**	0.0446	**India**	0.0229	0.2084
10	Tunisia	0.0219*	0.0001	Hungary	0.0214	0.0751	Indonesia	0.0433	**Colobmia**	0.0215	0.3507

Table A.7. High Excess *Valex* Return Countries in Different Periods

Countries with Lowest Excess *Valex* Return: $r^{A,Valex} - r^{L,Valex}$

Back

Rank	1991-2020	$r^{A,Vale} - r^{L,Vale}$	$p - val$	1991-2006	$r^{A,Vale} - r^{L,Vale}$	$p - val$	2007-2008	$r^{A,Vale} - r^{L,Vale}$	2010-2019	$r^{A,Vale} - r^{L,Vale}$	$p - val$
1	**Singapore**	-0.005	0.3405	Czech Republic	-0.008	0.5743	Japan	-0.070	Thailand	-0.011	0.3855
2	Czech Republic	-0.005	0.5215	**Singapore**	-0.004	0.6594	China	-0.046	Israel	-0.008	0.4735
3	**New Zealand**	-0.003	0.6403	Greece	-0.003	0.6186	Israel	-0.026	**New Zealand**	-0.007	0.2702
4	Greece	-0.002	0.5995	**New Zealand**	-0.003	0.7154	**Singapore**	-0.024	United States	-0.006	0.5049
5	Switzerland	-0.002	0.6599	Morocco	-0.003	0.6194	Egypt	-0.018	Switzerland	-0.006	0.1913
6	Israel	-0.001	0.7325	Pakistan	-0.000	0.9243	Switzerland	-0.015	Korea	-0.006	0.4538
7	Australia	-0.001	0.8098	Hong Kong SAR	0.0002	0.9498	Czech Republic	-0.015	Australia	-0.004	0.5939
8	Morocco	-0.000	0.8812	Austria	0.0010	0.6377	Peru	-0.011	**Singapore**	-0.004	0.4244
9	Japan	-0.000	0.9699	Denmark	0.0016	0.7135	Spain	-0.010	China	-0.002	0.7172
10	Austria	-1.404	0.9917	Finland	0.0017	0.7485	Morocco	-0.009	Spain	-0.001	0.7196

Table A.8. Low Excess *Valex* Return Countries in Different Periods

Regressions from Gourinchas and Rey (2022) Back

Table 4	$nagdp$	$vagdp$		$nagdp$	$vagdp$
1962:2-2015:4			1990:1-2015:4		
vix	-0.60**	-0.05**		-0.50**	-0.09**
	(.11)	(.02)		(.09)	(.03)
c	-1.75	1.28**		-1.75	2.52
	(2.1)	(.36)		(2.1)	(.70)
N	190	190		80	80
Adj. R^2	0.14	0.04		0.26	0.11

Table 4: Exorbitant Duty over Time. The table reports the results from an OLS regression of the U.S. net foreign asset position relative to GDP ($nagdp$) on the VIX index extended before 1986 with the volatility of the MSCI-ex US index. $vagdp$ refers to the valuation component (relative to GDP) defined as $VA_t = NA_t - NAt - 1 - FX_t$ where FA_t represents the net financial flows in period t.

Fig A.1. Table 4 in Gourinchas and Rey (2022)

Definition of Uncertainty Measures

Back

- **Definitions of UI Measures**

1) GZ Spread: Unweighted average of secondary market prices of senior unsecured bonds issued by a large representative sample of U.S. non-financial firms. (May try EBP later.)
2) World Uncertainty Index: GDP weigthed average of country specific uncertainty index based on the quarterly Economist Intelligence Unit (EIU) country reports.
3) Financial Stress Index: A newspaper-based Financial Stress Indicator (FSI) for the United States developed by Lukas Püttmann.
4) Policy Uncertainty Index: 10 Large newspapers (USA Today, the Miami Herald, the Chicago Tribune, the Washington Post, the Los Angeles Times, the Boston Globe, the San Francisco Chronicle, the Dallas Morning News, the Houston Chronicle, and the WSJ) based economic and policy uncertainty.
5) VIX: Real-time index that represents the market' s expectations for the relative strength of near-term price changes of the *SP* 500 Index (SPX).
6) Global Financial Cycles: Miranda-Agrippino and Rey (2022)
7) BAA Spread: Spread between Moody's Seasoned Baa Corporate Bond and 10-Year Treasury

The Other Side of the Ledger of the US's Exorbitant Duty

Back

→ β for each uni-variate regression. (R^2: 0.00∼0.03)

	non-US (1990-2020)	ADV (1990-2020)	EME (1990-2020)	non-US (1990-2006)	non-US (2010-2020)
No. Observations	1433	718	715	747	490
$\Delta(-)GFC$	0.015	0.013	0.017	0.037***	0.016
	(1.278)	(1.595)	(0.935)	(2.604)	(1.228)
$\Delta ln(VIX)$	0.042*	0.024	0.060**	0.087**	0.051
	(1.770)	(0.802)	(1.977)	(2.322)	(1.172)
$\Delta ln(BAA)$	0.044**	0.026	0.062**	0.051	0.070***
	(2.125)	(1.191)	(1.991)	(0.888)	(3.187)
$\Delta ln(PU)$	0.046***	0.068***	0.025	0.102**	0.019
	(2.491)	(2.505)	(1.008)	(2.071)	(0.977)
$\Delta ln(FS)$	2.959***	2.100*	3.839***	3.387*	5.627***
	(2.505)	(1.950)	(2.451)	(1.910)	(16.125)
$\Delta ln(GZ)$	0.040***	0.010	0.070***	0.045	0.058***
	(2.355)	(0.557)	(2.749)	(0.974)	(8.373)
$\Delta ln(WUI)$	-0.009	-0.030	-0.018	0.032	-0.014
	(-0.359)	(-0.012)	(-0.625)	(1.223)	(-0.391)

Law of Motion for NFA/GDP Back

$$\frac{\Delta NFA_t}{GDP_t} \simeq \frac{TB_t}{GDP_t} + \frac{(II_t^A - II_t^L) + (V_t^A - V_t^L)}{GDP_t}$$

$$= \frac{TB_t}{GDP_t} + \underbrace{\overbrace{\frac{1}{2}\frac{(A_{t-1} + L_{t-1})}{GDP_t}}^{\text{Gross External Position}} \underbrace{(r_t^A - r_t^L)}_{\text{Excess Return}}}_{\text{Return Effect}} + \underbrace{\overbrace{\frac{1}{2}(r_t^A + r_t^L)} \underbrace{\frac{(A_{t-1} - L_{t-1})}{GDP_t}}_{NFA_{t-1}/GDP_t}}_{\text{Position Effect}}$$

$$= \frac{TB_t}{GDP_t} + \underbrace{\overbrace{\frac{1}{2}\frac{(A_{t-1} + L_{t-1})}{GDP_t}}^{\text{Gross External Position}} \underbrace{(r_t^{A,II} - r_t^{L,II}) + (r_t^{A,Valoth} - r_t^{L,Valoth}) + (r_t^{A,Valex} - r_t^{L,Valex})}_{\text{Excess Return}}}_{\text{Return Effect}} + \underbrace{\overbrace{\frac{1}{2}(r_t^A + r_t^L)} \underbrace{\frac{(A_{t-1} - L_{t-1})}{GDP_t}}_{NFA_{t-1}/GDP_t}}_{\text{Position Effect}}$$

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